

Risk Register

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
Paper triage forms missing vital information before AI assessment	Operational	High	High	Require completion of essential fields (chief complaint, vital signs, pain score and history) before the AI generates an ESI recommendation, with automatic missing-data alerts.	Missing required triage fields fall below 5%, and AI recommendations are successfully generated for over 95% of patients.
Nurses accepting AI-generated ESI scores without applying clinical judgement during busy	Ethical	Medium	High	Keep the AI advisory only, require nurses to confirm or modify every recommendation, and review overridden cases during	Audit results show appropriate nurse override rates and no increase in inappropriate AI-only decisions.

Mercer shifts				monthly quality meetings.	
Alert fatigue from repeated deterioration alerts for patients waiting in the Mercer ED waiting room	Operational	Medium	Medium	Configure deterioration alerts to trigger only after clinically significant changes in waiting time or repeat vital signs, while suppressing duplicate alerts.	More than 90% of high-priority alerts receive nurse review, and unnecessary repeat alerts decrease over time.
Real-time Mercer ED workflow changes reduce AI prediction accuracy over time (model drift)	AI-Technical	Medium	High	Monitor model performance monthly and retrain the model using newly collected Mercer ED patient data when accuracy declines.	Model accuracy remains within 5% of its original validation performance over time.
Algorithm performs worse for under-represent	Ethical / Equity	Medium	High	Conduct bias audits by demographic group before	Difference in sensitivity between demographic

ed patient groups (e.g., elderly, darker skin tones, rural referrals, atypical presentations)				deployment and after every model update. Retrain if performance gaps exceed predefined thresholds.	groups remains below 5%.
Model trained on historical Mercer data inherits existing triage bias	AI-Technical	Medium	High	Review historical labels, perform fairness testing, include clinician review during model development.	External validation demonstrates equal or improved performance compared with standard ESI.
Staff resistance and poor adoption of AI recommendations	Operational	Medium	Medium	Nurse education, phased rollout, clinician feedback sessions, maintain AI as decision support only.	>90% routine use with positive staff satisfaction scores.
Patient privacy or confidential information	Structural / Governance	Low	High	De-identify training data, role-based	Zero reportable privacy breaches during

exposed during AI processing				access, audit logs, secure storage.	evaluation period.
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